AHT Assignment

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(i) given that

$H = 20\ m$

$T = 65°C$

$T_\infty = 25°C$

Stagnant air at 1 atm $= P_\infty$

We need to compute Grashof Number Gr_H, to determine dominance of Natural convection over forced convection

$$Gr_H = \frac{g \beta \Delta T H^3}{\nu^2} \qquad g = 9.81\ m/s^2$$

β = Volumetric thermal expansion coefficient $= \frac{1}{T_\infty} = 0.00335\ K^{-1}$

$\Delta T = 40K$

$H = 20m$

ν = Kinematic viscosity of air at $T_\infty = 1.562 \times 10^{-5}\ m^2/s$

So, $$Gr_H = \frac{9.81 \times 0.00335 \times 40 \times 20^3}{1.562^2 \times 10^{-10}}$$

$$= 4.31025 \times 10^{13}$$

Clearly, this extremely large value indicates that buoyant forces dominate over viscous force in the system.

Hence Natural convection dominates over forced convection in the given problem

(ii) Momentum equation $\rightarrow$ $$u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = \nu \frac{\partial^2 v}{\partial x^2} + g\beta(T - T_\infty)$$

Energy equation $\rightarrow$ $$u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial x^2}$$

Continuity equation $\rightarrow$ $$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

Assuming a linear gradient of temperature in x-direction and no temperature change in y-direction

We can see streamline & heatline plots by numerically solving

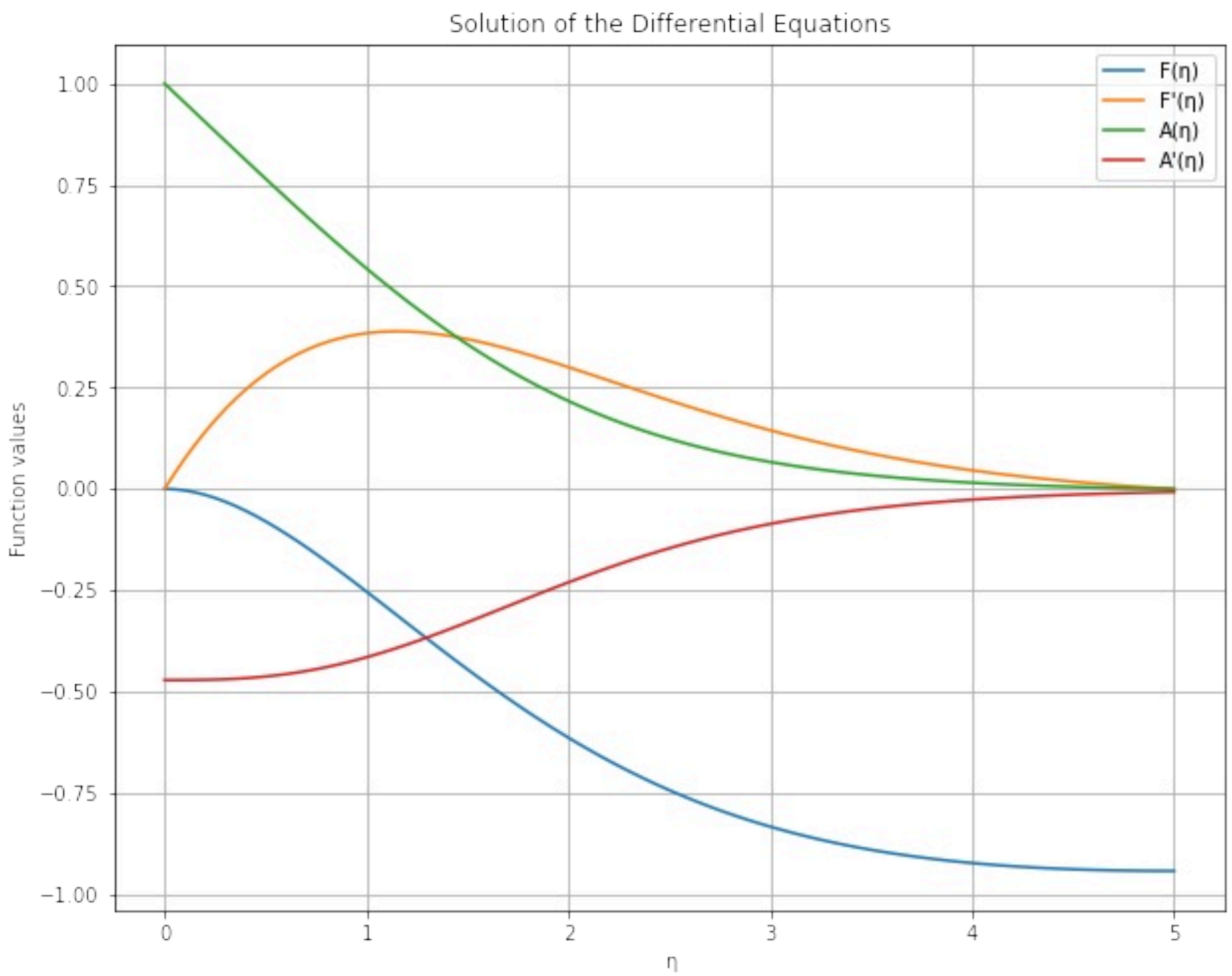
Using code we can see that,

$\bar{\delta}_T = 14.38$ cm $\qquad \overline{Nu}_{0-H} = 1390.77186$

Thermal boundary layer thickness

Heat Transfer coefficient, $\bar{h} = 1.8247\ W/m^2K$

Heat flux per unit width $= 72.98771\ W/m^2$



(iii) Obtaining Average Nusselt Number $\overline{Nu}_H$ from similarity solution

$$\overline{Nu}_H = 0.59\, Gr_H^{1/4}\, Pr^{1/4}$$

where Gr_H = Grashof Number $= 4.31025 \times 10^{13}$

Pr_H = Prandtl Number $= 0.697633$

Thus, $\overline{Nu}_H = 1381.9018$

Average Thermal Boundary layer thickness $\delta_{th} = \frac{H}{\overline{Nu}_H}$

$\delta_{th} = 14.47\,cm$

Heat Transfer coefficient, $h = \frac{\overline{Nu}_H k}{H} = 1.8130552 \frac{W}{m^2 K}$

Heat flux, $q'' = h(T_0 - T_\infty) = 72.5222\ W/m^2$

Code attached